

Section : Common

Q.1  $\int \frac{(x^4 - x)^{1/4}}{x^5} dx$  is equal to

1.  $\frac{4}{15} \left(1 - \frac{1}{x^3}\right)^{5/4} + C$  :  $C$  is a constant of integration
2.  $\frac{4}{3} \left(1 - \frac{1}{x^3}\right)^{5/4} + C$  :  $C$  is a constant of integration
3.  $\frac{4}{15} \left(1 - \frac{1}{x^3}\right)^{3/4} + C$  :  $C$  is a constant of integration
4.  $\frac{4}{15} \left(1 - \frac{1}{x^3}\right)^{1/5} + C$  :  $C$  is a constant of integration

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.2 The real valued function  $f(x) = 12x^{\frac{4}{3}} - 6x^{\frac{1}{3}}$ ,  $x \in [-8, 8]$  has absolute maximum value equal to

1. 0
2.  $-\frac{9}{4}$
3. 180
4. 204

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.3 If  $A$  is square matrix of order  $3 \times 3$  and  $|\text{adj } A| = 64$ , then the value of  $|5A|$  is

1. 100
2.  $\pm 1000$
3. 320
4.  $\pm 320$

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.4 The area of the region enclosed between the parabola  $y = \frac{3x^2}{4}$  and the line  $3x - 2y = 12$  is,

1. 63 sq. units
2. 36 sq. units
3. 27 sq. units
4. 18 sq. units

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.5 Match List-I with List-II

List-I  
Differential equation

List-II  
Order and degree

(A)  $(y'')^3 + (y')^4 - 6 = (y''')^2$  (I) Order = 1, Degree = 2

(B)  $\sqrt{(y')^2 + 5} = y''$  (II) Order = 2, Degree = 3

(C)  $(y')^2 = (2 + y'')^{3/2}$  (III) Order = 2, Degree = 2

(D)  $y = xy' + \sqrt{a^2(y')^2 + b^2}$  (IV) Order = 3, Degree = 2

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

2. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)

3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

4. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.6 If the interval in which the function  $f(x) = 4x^3 - 6x^2 - 72x + 30$  is strictly decreasing, is (a,b) then a+b is equal to

1.  $\infty$

2. 0

3. 1

4. 2

Options 1. 1

2. 2

3. 3

4. 4

Q.7 For the given linear programming problem  $z = ax + by$ ,  $a, b > 0$  subject to the constraints

$2x + y \leq 10$ ,  $x + 3y \leq 15$ ,  $x, y \geq 0$ . If the corner points are (0, 0), (5, 0), (3, 4) and (0, 5)

and  $z$  is maximum at both (3, 4) and (0, 5), then the relationship between  $a$  and  $b$  is

1.  $b = 3a$

2.  $a = 3b$

3.  $a = 2b$

4.  $a = b$

Options 1. 1

2. 2

3. 3

4. 4

Q.8 For  $x > y > 0$ , if  $x^5 y^6 = (x + y)^{11}$ , then  $\frac{d^2 y}{dx^2}$  is

1.  $x/y$

2.  $y/x$

3. 1

4. 0

Options 1. 1

2. 2

3. 3

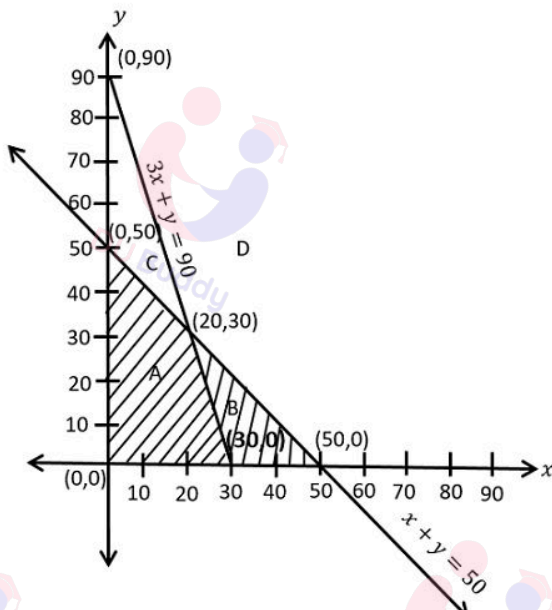
Q.9 If A is a matrix of order  $m \times n$  and B is a matrix such that  $AB^T$  and  $B^T A$  are both well-defined matrices, then order of matrix B is

1.  $n \times n$
2.  $m \times m$
3.  $m \times n$
4.  $n \times m$

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.10

The feasible region represented by the constraints  $x + y \leq 50$ ,  $3x + y \leq 90$ ,  $x \geq 0$ ,  $y \geq 0$  of an LPP is



1. Region D
2. Region B
3. Region A
4. Region C

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.11

If  $\begin{bmatrix} x-2 & 3 & -2 \\ y & 0 & -4 \\ 2 & z & 0 \end{bmatrix}$  is a skew symmetric matrix, then the value of  $x + y + z$  is

1. 9
2. -1
3. 7
4. 3

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.12 Consider the differential equation  $xdy = (x + y)dx$ . Which of the following are true?

- (A) It is a homogenous differential equation

(B) It is a differential equation of order 2

(C) The general solution of the differential equation contains 2 arbitrary constants

(D) Integrating factor of differential equation is  $\frac{1}{x}$

(E) Degree of the differential equation is not defined

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only

2. (A), (D) and (E) only

3. (A) and (D) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.13  $\int \frac{f'(x)}{f(x) \log_e[f(x)]} dx$  is equal to

1.  $f(x) \cdot \log_e[f(x)] + C$  :  $C$  is a constant of integration

2.  $\frac{\log_e[f(x)]}{f(x)} + C$  :  $C$  is a constant of integration

3.  $\log_e(\log_e[f(x)]) + C$  :  $C$  is a constant of integration

4.  $\frac{\log_e(\log_e[f(x)])}{f(x)} + C$  :  $C$  is a constant of integration

Options 1. 1

2. 2

3. 3

4. 4

Q.14 If a random variable X has the following probability distribution:

|      |               |               |               |               |
|------|---------------|---------------|---------------|---------------|
| X    | 0             | 1             | 2             | 3             |
| P(X) | $\frac{K}{2}$ | $\frac{K}{4}$ | $\frac{K}{4}$ | $\frac{K}{8}$ |

then,

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}                      \textbf{List-II}

(A) The value of K is (I)  $\frac{2}{15}$

(B)  $P(0 < X < 2)$  is (II)  $\frac{1}{15}$

(C)  $P(1 < X < 3)$  is (III)  $\frac{8}{15}$

(D)  $P(X > 2)$  is (IV)  $\frac{4}{15}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

4. (A) - (IV), (B) - (I), (C) - (III), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.15 If  $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$  and  $2A^{-1} = KI - A$ , where K is a real number and I is the identity matrix of order 2, then the value of K is:

1. 5
2. -3
3. -6
4. 9

Options 1. 1  
2. 2  
3. 3  
4. 4

Section : Applied Mathematics

**Q.16** If the probability that an individual suffers a bad reaction from an injection of a given serum is 0.001, the probability that out of 2000 individuals, more than two individuals suffer from bad reaction.

[Given that  $e^{-2} = 0.13534$ ]

1. 0.6767
2. 0.3233
3. 0.18044
4. 0.9753

Options 1. 1  
2. 2  
3. 3  
4. 4

**Q.17** If  $X$  is a random variable and  $a, b$  are real numbers, then which of the following statements are correct?

- (A)  $E[aX + b] = aE(X) + b$
- (B)  $\text{Var}(aX + b) = a^2\text{Var}(X) + b$
- (C)  $\text{Var}(aX + b) = a\text{Var}(X)$
- (D)  $\text{Var}(X) = E(X^2) - [E(X)]^2$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (D) only
3. (B) and (C) only
4. (A), (C) and (D) only

Options 1. 1  
2. 2  
3. 3  
4. 4

**Q.18**

|                             |      |      |      |      |      |      |      |
|-----------------------------|------|------|------|------|------|------|------|
| Year(x)                     | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
| Sales<br>(in lakh Rs. ) (y) | 65   | 68   | 70   | 72   | 75   | 67   | 73   |

If  $y = 70 + 0.964(x - 2014)$  be a straight line trend fitted by the method of the least squares to the above data, then the trend value for the year 2015 is:

1. 70.96
2. 72.12
3. 76.12
4. 75.54

Options 1. 1  
2. 2  
3. 3  
4. 4

**Q.19**

A measure of an average yearly growth of an investment over a certain time period when returns are reinvested is

1. Compound annual growth rate
2. Annual average growth rate
3. Nominal rate
4. Linear depreciation rate

Options 1. 1  
2. 2  
3. 3  
4. 4

**Q.20** The solution set of  $6 \leq -3(2x - 4) < 12$ ,  $x \in \mathbb{R}$  is:

1.  $(0, 1]$
2.  $[1, 0)$

3.  $(0, 1)$

4.  $[0, 1]$

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Two pipes A and B can fill a tank in 20 minutes and 30 minutes respectively. Both pipes A and B are opened together for some time and then pipe B is turned off. If the tank is filled in 15 minutes, then after how many minutes is pipe B turned off?

1. 5 minutes

2. 8 minutes

3. 12 minutes

4. 7.5 minutes

Options 1. 1

2. 2

3. 3

4. 4

Q.22 The unit's digit of  $12^{12}$  is equal to:

1. 1

2. 4

3. 6

4. 2

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Which of the following are similarities between the sinking fund and the savings account?

(A) The sinking fund and the savings account are both financial tools.

(B) Both can be used in any emergency.

(C) They both involve setting aside an amount of money for the future.

(D) Both are long-term accounts which can be closed any time.

Choose the correct answer from the options given below

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (B), (C) and (D) only

4. (A) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.24 If A and B are symmetric matrices of the same order, then

1.  $(AB+BA)$  is skew symmetric matrix.

2.  $(AB+BA)$  is always a diagonal matrix.

3.  $(AB+BA)$  is a symmetric matrix.

4.  $(AB-BA)$  is a symmetric matrix.

Options 1. 1

2. 2

3. 3

4. 4

Q.25 If the objective function for a linear programming problem (LPP) is  $Z = 4x + 5y$  and the corner points of the bounded feasible region are  $(9, 0)$ ,  $(4, 3)$ ,  $(2, 5)$ , and  $(0, 8)$ , then the minimum value of Z is:

1. 0

2. 31

3. 36

4. 40

Options 1. 1

2. 2

3. 3

4. 4

Q.26 Which of the following is NOT correct?

1. Nominal rate of return is the total amount of money earned by an investment without considering various expenses.

2. The compound annual growth rate is calculated by the final and beginning value of the investment.

3. The compound annual growth rate describes the effect of compounding.

4. The actual rate of return can be calculated by adding the various expenses into the total money earned.

Options 1. 1

2. 2

3. 3

Q.27 List 1 List 2  
( Matrix A ) ( Determinant of adj A )

A (I) 6

$$\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix}$$

B (II) 5

$$\begin{bmatrix} 0 & 1 \\ 4 & -1 \end{bmatrix}$$

C (III) -4

$$\begin{bmatrix} 1 & 2 \\ -3 & -1 \end{bmatrix}$$

D (IV) -2

$$\begin{bmatrix} 4 & -2 \\ 3 & 0 \end{bmatrix}$$

Match List-I with List-II

Choose the correct answer from the options given below:

- (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
- (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
- (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
- (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Options 1. 1

2. 2
3. 3
4. 4

Q.28 If  $I = \int \frac{x}{x - \sqrt{x^2 - 4}} dx = \alpha x^3 + \beta(x^2 - 4)^{\frac{3}{2}} + \gamma$ ,

where  $\gamma$  is a constant of integration, then

1.  $\alpha = 2\beta$
2.  $2\alpha = \beta$
3.  $\alpha = \beta$
4.  $\alpha = 3\beta$

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Mr. X wishes to purchase a flat for Rs. 44, 60, 800 with a down payment of Rs. 10, 00, 000 and the balance in equal monthly instalments (EMI) for 20 years.

If the bank charges 7.5% per annum compounded monthly, then the EMI is:

[Given that  $(1.00625)^{240} = 4.4608$ ]

1. Rs. 31250
2. Rs. 37285
3. Rs. 27880
4. Rs. 30280

Options 1. 1

2. 2
3. 3
4. 4

Q.30 A T.V. panel costing Rs 18000 has a useful life of 12 years. If the annual depreciation is Rs 1000, then its scrap value by linear method is:

1. Rs. 8000
2. Rs. 6000
3. Rs. 5000
4. Rs. 7500

Options 1. 1

2. 2
3. 3
4. 4

Q.31 A fisherman is rowing a boat. He takes 6 hours to row 48 km upstream whereas he takes 3 hours to go to the same distance downstream.

Based on the above information, which of the following are correct?

- His speed of rowing in still water is 10 km/hour.
- The speed of the stream is 3 km/hour.
- The average speed of the fisherman is  $32/3$  km/hour.
- The speed of the stream is 4 km/hour.

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (C) and (D) only

3. (A), (B) and (C) only  
4. (A) and (C) only

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.32

The value of the definite integral  $I = \int_{-1}^1 \frac{1}{1 + \sqrt{e^x}} dx$  is:

1. 0  
2. 1  
3. 2  
4. 3

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.33

The general solution of the differential equation

$$(x^2 - yx^2) dy + (y^2 + x^2y^2) dx = 0 \text{ is:}$$

1.  $\log_e |y| + \frac{1}{x} + \frac{1}{y} - x = c$   
2.  $\log_e |y| - \frac{1}{x} + \frac{1}{y} + x = c$   
3.  $\log_e |x| - \frac{1}{x} + \frac{1}{y} + x = c$   
4.  $\log_e |x| + \frac{1}{x} + \frac{1}{y} + x = c$

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.34

If  $e^y(x+1) = 1$ , then

1.  $\frac{d^2y}{dx^2} = 2 \frac{dy}{dx}$   
2.  $\frac{d^2y}{dx^2} = 2 \left( \frac{dy}{dx} \right)^2$   
3.  $\frac{d^2y}{dx^2} = 2 \left( \frac{dy}{dx} \right)^{1/2}$   
4.  $\frac{d^2y}{dx^2} = \left( \frac{dy}{dx} \right)^2$

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.35

If a random variable  $X$  has the following probability distribution:

X    0   1   2   3   4

P(X) k   2k   3k    $k^2$     $6k^2$

then match List-I with List-II

List 1

List 2

- (A) k                      (I)  $3/7$   
(B)  $P(X < 2)$             (II)  $6/49$   
(C)  $P(X > 3)$             (III)  $1/7$   
(D)  $P(2 \leq X \leq 3)$       (IV)  $22/49$

Choose the correct answer from the options given below:

1. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)  
2. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)  
3. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)  
4. (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.36

$$\text{If } \begin{bmatrix} 1 & 0 \\ b & 5 \end{bmatrix} + 2 \begin{bmatrix} a & 0 \\ 1 & -2 \end{bmatrix} = I,$$

where  $I$  is the unit matrix of order 2, then the value of  $(a - b)$  is:

1. 0
2. 1
3. 2
4. 3

Options 1. 1

2. 2
3. 3
4. 4

Q.37

A person has invested Rs. 20,000 in 2020 for 5 years.

If the CAGR for his investment is 11.84%, the end balance of his investment is

[Given  $(1.1184)^5 = 1.7498$ ]

1. Rs. 35998
2. Rs. 34996
3. Rs. 33998
4. Rs. 34448

Options 1. 1

2. 2
3. 3
4. 4

Q.38 In reference to the Inferential Statistics, which of the following is NOT correct?

1. A statistical hypothesis is a statement about the population parameter which may be true or false.
2. The standard deviation of the sampling distribution of a statistics is known as its standard error.
3. If sufficiently large random samples with replacement are drawn from a population, then the sampling distribution of the sample means approaches a binomial distribution.
4. In a statistical hypothesis testing, a Type-I error is made when null hypothesis is rejected when it is true.

Options 1. 1

2. 2
3. 3
4. 4

Q.39

The number of letters posted in a certain city on each day for a week is given as follows:

40, 55, 28, 25, 31, 52, 43

Which of the following is NOT an entry in the three-day moving averages?

1. 41
2. 36
3. 42
4. 37

Options 1. 1

2. 2
3. 3
4. 4

Q.40

If the corner points of the bounded feasible region for a linear programming problem (LPP) are  $(0, 2)$ ,  $(3, 0)$ ,  $(6, 0)$ ,  $(6, 8)$  and  $(0, 5)$ , then which of the following are correct for the objective function  $Z = 4x + 6y$ ?

- (A) The minimum value of the objective function occurs at  $(0, 2)$  and  $(3, 0)$  only.
- (B) The minimum value of the objective function occurs at the mid-point of the line segment joining the points  $(0, 2)$  and  $(3, 0)$  only.
- (C) The minimum value of the objective function occurs at every point of the line segment joining the points  $(0, 2)$  and  $(3, 0)$ .
- (D) The difference between the maximum value and minimum value of the objective function is 60.

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A) and (C) only
3. (B) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.41

The pattern and behavior of the data in any time series are based on which of the following components:

- (A) Secular trend component
- (B) Seasonal component
- (C) Cyclical component
- (D) Irregular component

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.42

Let two independent random samples of sizes  $n_1$  and  $n_2$  respectively have been drawn from the same normal population. Let  $\bar{X}_1$  and  $\bar{X}_2$  be the means and let  $s_1$  and  $s_2$  be their standard deviations. In order to test whether the two sample means  $\bar{X}_1$  and  $\bar{X}_2$  differ significantly or not, the  $t$ -test statistic is given by

1.  $t = \frac{\bar{X}_1 - \bar{X}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ ,  $S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$
2.  $t = \frac{\bar{X}_1 - \bar{X}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ ,  $S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$
3.  $t = \frac{\bar{X}_1 - \bar{X}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ ,  $S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$
4.  $t = \frac{\bar{X}_1 - \bar{X}_2}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ ,  $S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.43

A die is tossed 6 times and getting 1 or 5 is considered a success. The probability of getting at least one success in six tosses is:

1.  $\frac{1}{3}$
2.  $\frac{1}{729}$
3.  $\frac{728}{729}$
4.  $\frac{665}{729}$

Options 1. 1

2. 2
3. 3
4. 4

Q.44

Let  $A = [a_{ij}]$  be a square matrix, where  $a_{ij} = \begin{cases} 0, & \text{when } i = j \\ 1, & \text{otherwise} \end{cases}$   
If  $|\text{adj } A| = |A|^2$ , then which of the following statements are correct?

- (A)  $A$  is a skew-symmetric matrix.
- (B)  $A$  is a non-singular matrix.
- (C)  $A$  is a square matrix of order 4.
- (D)  $A$  is a symmetric matrix.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B) and (D) only
3. (A) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.45

The average salary per head of the entire staff of a small factory including managers and technicians is Rs. 5750.

The average salary per head of a manager is Rs. 20000

and that of a technician is Rs. 5000.

If there are 4 managers, then the number of technicians in the factory is:

1. 48
2. 64
3. 76
4. 72

Options 1. 1

2. 2
3. 3
4. 4

Q.46

If  $A = [a_{ij}]$  is a square matrix of order 3 such that  $a_{ij} = i + j$  for all  $i, j$ , then which of the following are correct?

- (A)  $A$  is a skew-symmetric matrix.

(B)  $A$  is a non-singular matrix.

(C) The inverse of  $A$  does not exist.

(D)  $A$  is a symmetric matrix.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B) and (D) only
3. (A) and (C) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.47 In which of the following intervals, the function  $f(x) = \frac{x}{\log x}$  is decreasing?

1.  $(-\infty, e)$
2.  $(0, e)$
3.  $(0, e) - \{1\}$
4.  $(e, \infty)$

Options 1. 1

2. 2
3. 3
4. 4

Q.48 The equation of the tangent to the curve  $y = \frac{x-3}{(x-1)(x-2)}$  at the point where it cuts the  $x$ -axis is:

1.  $x + 2y - 3 = 0$
2.  $2y - x - 3 = 0$
3.  $2y - x + 3 = 0$
4.  $3x + 2y - 3 = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.49 A company has been producing steel tubes of mean inner diameter 2 cm. A sample of 10 tubes gives a mean inner diameter of 2.01 cm and a variance of  $0.0004 \text{ cm}^2$ .

The value of the test statistic is:

1. 2.5
2. 1.5
3. 2.36
4. 0.512

Options 1. 1

2. 2
3. 3
4. 4

Q.50 In a 10 km race,  $A$ ,  $B$  and  $C$  each run at uniform speed and get the first, second and third positions respectively. If  $A$  beats  $B$  by 1 km and  $B$  beats  $C$  by 1 km, then by how much distance does  $A$  beat  $C$ ?

1. 2 km
2. 1.9 km
3. 1.8 km
4. 2.1 km

Options 1. 1

2. 2
3. 3
4. 4

Section : Core Mathematics

Q.51 Which of the following statements are correct?

(A) If  $\vec{a}$  and  $\vec{b}$  represent the adjacent sides of a triangle, then its area is  $\frac{1}{2} |\vec{a} \times \vec{b}|$

(B) If  $\vec{a}$  and  $\vec{b}$  represent the adjacent sides of a parallelogram, then its area is  $|\vec{a} \times \vec{b}|$

(C)  $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \cos \theta$

(D) If  $\vec{a}$  and  $\vec{b}$  represent the 'diagonals' of a parallelogram, then its area is  $\frac{1}{2}|\vec{a} \times \vec{b}|$

Choose the correct answer from the options given below:

1. (B), (C) and (D) only

2. (C) and (D) only

3. (A), (B) and (D) only

4. (A) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.52

If  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  are three unit vectors such that  $\vec{a} + 2\vec{b} - 3\vec{c} = \vec{0}$ , then the value of  $2\vec{a} \cdot \vec{b} - 6\vec{b} \cdot \vec{c} - 3\vec{c} \cdot \vec{a}$  is

1. -7

2. -14

3. 7

4. 14

Options 1. 1

2. 2

3. 3

4. 4

Q.53

The area bounded by the curve  $y = \log x$ ,  $y = 0$  and  $x = e$ , is

1.  $e - 1$  sq. units

2.  $e$  sq. units

3. 1 sq. units

4. 0 sq. units

Options 1. 1

2. 2

3. 3

4. 4

Q.54

The value of  $k$  for which the function, defined by,  $f(x) = \begin{cases} \frac{3x+4\tan x}{x} & : x \neq 0 \\ k & : x = 0 \end{cases}$  is continuous at  $x = 0$ , is

1. 3

2. 4

3. 7

4. 0

Options 1. 1

2. 2

3. 3

4. 4

Q.55

Consider the following L.P.P minimize  $z = x - 7y + 190$  subject to  $x + y \leq 8$ ,  $x + y \geq 4$ ,  $x \leq 5$ ,  $y \leq 5$  and  $x, y \geq 0$ . Then which of the following is/are true?

(A) It's feasible region is unbounded

(B) It's feasible region is bounded

(C) It's feasible region has 5 corner points

(D) It's feasible region has 6 corner points

Choose the correct answer from the options given below:

1. (A) only
2. (B) and (C) only
3. (A) and (C) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

**Q.56** The area of the region bounded by  $y = -1$ ,  $y = 2$ ,  $x = y^3$  and  $x = 0$  is  $\frac{m}{n}$  sq. units, where  $\gcd(m, n) = 1$ , then  $m - n$  is

equal to

1. 9
2. 11
3. 13
4. 15

Options 1. 1

2. 2
3. 3
4. 4

**Q.57** The domain of the function  $\cos^{-1}(2x - 3)$  is

1.  $[1, 2]$
2.  $[-1, 1]$
3.  $(1, 2)$
4.  $(-1, 1)$

Options 1. 1

2. 2
3. 3
4. 4

**Q.58** Which of the following functions  $f(x)$  are differentiable at  $x = 0$ ?

- (A)  $|x|$
- (B)  $|x - 1|$
- (C)  $|\sin x|$
- (D)  $|\cos x|$
- (E)  $x^2$

Choose the \textbf{correct} answer from the options given below:

1. (A), (C) and (D) only
2. (B) and (E) only
3. (B), (D) and (E) only
4. (A), (C) and (E) only

Options 1. 1

2. 2
3. 3
4. 4

**Q.59** Consider the curve which is represented by the differential equation  $\frac{dy}{dx} = 1 + x + y + xy$ . If it passes through the point  $(0, 0)$ , then which of the following is/are true?

- (A) it is a straight line.

(B) it is a parabola.

(C) it also passes through the point  $\left(-1, \frac{1}{\sqrt{e}} - 1\right)$

(D) Its equation is  $xy(x+1)\left(y - \frac{1}{\sqrt{e}} + 1\right) = 0$

Choose the \textbf{correct} answer from the options given below:

1. (C) and (D) only

2. (A) only

3. (B) only

4. (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.60

If  $\begin{vmatrix} -a^2 & ab & ac \\ ba & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = k \cdot a^l b^m c^n$ , then

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \quad \textbf{List-II}

(A)  $l = m = n =$  (I) 10

(B)  $k + l + m + n =$  (II) 6

(C)  $k^2 + l^2 + (m - n)^2 =$  (III) 2

(D)  $l^2 + m^2 + (n - k) =$  (IV) 20

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

3. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.61

The value of  $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin |x| + \cos |x|) dx$  is

1. 0

2. 2

3. 4

4.  $\frac{5}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.62

If  $\hat{a}$ ,  $\hat{b}$  and  $\hat{c}$  are three units vectors and  $\hat{a} + \hat{b} + \hat{c} = \vec{0}$ , then the angle between  $\hat{a}$  and  $(-\hat{b})$  is

1.  $\frac{2\pi}{3}$

2.  $\frac{\pi}{3}$

3.  $\frac{\pi}{2}$

4.  $\frac{\pi}{6}$

Options 1. 1

2. 2

3. 3

4. 4

Q.63 Two events A and B will be independent, then

1.  $P(\bar{A} \cap \bar{B}) = [1 - P(A)][1 - P(B)]$

2.  $P(A) + P(B) = 1$

3.  $P(A) = P(B)$

4.  $P(A) + P(B) = 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.64 If  $A = \{1, 2, 3, 4, \dots, n\}$  and  $B = \{x, y\}$ , then the number of surjections from A to B is

1.  $2^n - 1$

2.  $2^n - 2$

3.  ${}^nP_2$

4.  $n!$

Options 1. 1

2. 2

3. 3

4. 4

Q.65 If A and B are two square symmetric matrices of same order, then AB-BA is

1. a symmetric matrix

2. a skew-symmetric matrix

3. neither a symmetric nor a skew-symmetric

4. always a diagonal matrix

Options 1. 1

2. 2

3. 3

4. 4

Q.66 If the vectors  $\vec{a} = 3\hat{i} - p\hat{j} + 5\hat{k}$  and  $\vec{b} = -6\hat{i} + 14\hat{j} + q\hat{k}$  are collinear, then the value of p and q are:

1.  $p = -7, q = 18$

2.  $p = 7, q = -20$

3.  $p = 7, q = -10$

4.  $p = -7, q = 5$

Options 1. 1

2. 2

3. 3

4. 4

Q.67

The co-ordinates of the point at which the line  $\frac{x-3}{3} = \frac{y+1}{2} = \frac{z-4}{-2}$  crosses  $xy$  plane, are

1. (6, 1, 0)
2. (-3, -5, 0)
3. (9, 3, 0)
4. (0, -3, 0)

Options 1. 1

2. 2
3. 3
4. 4

Q.68

An objective function  $z = ax + by$  is maximum at points (15, 15) and (0, 20). If  $a, b \geq 0$  and  $ab = 27$ , then the maximum value of the objective function is

1. 190
2. 180
3. 170
4. 160

Options 1. 1

2. 2
3. 3
4. 4

Q.69

A relation R in the set  $A = \{1, 2, 3, 4\}$  is given by  $R = \{(1, 1), (2, 2), (1, 2), (2, 3), (3, 4), (4, 4), (1, 3), (2, 4), (1, 4)\}$  is

1. Reflexive
2. Symmetric
3. Transitive
4. Reflexive and transitive

Options 1. 1

2. 2
3. 3
4. 4

Q.70

If  $A = \begin{bmatrix} 5 & 3 \\ 2 & 4 \end{bmatrix}$  then the matrix  $A^2 - 6A + 14I$  (where I is an identity matrix of order 2)

1.  $\begin{bmatrix} 15 & 6 \\ 9 & 8 \end{bmatrix}$
2.  $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
3.  $\begin{bmatrix} 15 & 9 \\ -12 & -6 \end{bmatrix}$
4.  $\begin{bmatrix} 15 & 9 \\ 6 & 12 \end{bmatrix}$

Options 1. 1

2. 2
3. 3
4. 4

Q.71

The following system of equations  $2x - y + 3z = 5, 3x + 2y - z = 7, 4x + 5y - \lambda z = \mu$  is consistent. Then

1.  $\lambda = 5$  and  $\mu \neq 9$
2.  $\lambda \neq 9$  and  $\mu = 5$

3.  $\lambda = 5$  and  $\mu = 9$

4.  $\lambda = 5$  and  $\mu = 5$

Options 1. 1

2. 2

3. 3

4. 4

Q.72 If A speaks truth in 75\% cases and B speaks truth in 80\% cases, then the probability that they contradict each other in a statement, is:

1.  $\frac{13}{20}$

2.  $\frac{11}{20}$

3.  $\frac{7}{20}$

4.  $\frac{9}{20}$

Options 1. 1

2. 2

3. 3

4. 4

Q.73 The sides of an equilateral triangle are increasing at the rate of 2 cm/sec. The rate at which the area increases when the side is 10 cm, is

1.  $20\sqrt{3} \text{ cm}^2/\text{sec}$

2.  $10\sqrt{3} \text{ cm}^2/\text{sec}$

3.  $3\sqrt{5} \text{ cm}^2/\text{sec}$

4.  $5\sqrt{3} \text{ cm}^2/\text{sec}$

Options 1. 1

2. 2

3. 3

4. 4

Q.74 If A and B are two events such that  $P(A) = \frac{1}{2}$ ,  $P(B) = \frac{1}{3}$ ,  $P(B|A) = \frac{1}{4}$ , then  $P(A|B)$  is:

1.  $\frac{1}{8}$

2.  $\frac{7}{8}$

3.  $\frac{3}{8}$

4.  $\frac{5}{8}$

Options 1. 1

2. 2

3. 3

4. 4

Q.75 Match \textbf{List-I} with \textbf{List-II}

[.] denotes the greatest integer function.

**List-I**                      **List-II**

(A)  $\int_0^3 [x] dx$  (I)  $\frac{1}{2}$

(B)  $\int_0^1 [2x] dx$  (II) 1

(C)  $\int_0^1 [3x] dx$  (III)  $\frac{3}{2}$

(D)  $\int_0^1 [4x] dx$  (IV) 3

Choose the \textbf{correct} answer from the options given below:

- (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
- (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.76 Consider the equation of the line  $\vec{r} = -\hat{i} + 2\hat{k} + \mu(4\hat{i} - \hat{j} + 2\hat{k})$ .

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \quad \textbf{List-II}

- (A) It passes through the point (I) 4, -1, 2
- (B) Its direction ratios are (II)  $\frac{4}{\sqrt{21}}, \frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}$
- (C) Its Cartesian form is (III) (-1, 0, 2)
- (D) Its direction cosines are (IV)  $\frac{x+1}{4} = \frac{y}{-1} = \frac{z-2}{2}$

Choose the \textbf{correct} answer from the options given below:

- (A) - (I), (B) - (III), (C) - (IV), (D) - (II)
- (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.77 If  $A = \begin{bmatrix} 2 & 0 & 3 \\ -1 & 1 & 3 \\ 0 & -4 & 0 \end{bmatrix}$ , then the value of  $\det(2A)$  is

- 72
- 144
- 288
- 144

Options 1. 1  
2. 2  
3. 3  
4. 4

Q.78 If  $x = a \cos \alpha + b \sin \alpha$  and  $y = a \sin \alpha - b \cos \alpha$ , then  $\left(x \frac{dy}{dx} - y^2 \frac{d^2y}{dx^2}\right)$  is equal to:

- 0
- 2

4.  $y$

2.2

### 3.3

4.4

**Q.79** Match **List-I** with **List-II**

The function  $f(x) = (x - 1)(x + 1)^2$  has

**List-II**

(I)  $\frac{1}{3}$

(II) 0

(III) - 1

$$-(IV) = \frac{32}{27}$$

Choose the **correct** answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
2. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

2.2

### 3.3

4.4

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**Q.80** A player participates in 3 matches against three teams  $T_1, T_2$  and  $T_3$ . The probability of winning a match against teams

$T_1, T_2$  and  $T_3$  are 0.2, 0.3 and 0.9 respectively. If 'wins' can be regarded as independent events, then the probability that he

(A) wins all the 3 matches is 0.054

(B) wins no match is 0.054

(C) wins exactly two matches is 0.348

(D) wins exactly one match is 0.542

Choose the **correct** answer from the options given below:

1. (A), (C) and (D) only
2. (B), (C) and (D) only
3. (A) and (C) only
4. (A) and (D) only

2.2

### 3.3

4.4

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**Q.81** The general solution of the differential equation  $\frac{xdy}{dx} + 4y = x^3$ , ( $x \neq 0$ ) is:

1.  $y = \frac{x^3}{7} + cx^{-4}$ , c is arbitrary constant.

2.  $y = \frac{x^3}{7} + cx^{-7}$ ,  $c$  is arbitrary constant.

3.  $y = \frac{x^7}{7} + cx^{-4}$ ,  $c$  is arbitrary constant.

4.  $y = \frac{x^7}{7} + cx^{-3}$ ,  $c$  is arbitrary constant.

Options 1. 1

2. 2

3. 3

4. 4

Q.82

If the lines  $\frac{x-5}{7} = \frac{y+2}{-5} = \frac{z}{\lambda}$  and  $\frac{x}{1} = \frac{y}{2\lambda} = \frac{z}{3}$  are perpendicular to each other, then  $\lambda$  is equal to

1. 1

2. 2

3. 3

4. 4

Options 1. 1

2. 2

3. 3

4. 4

Q.83

If  $A = \begin{bmatrix} 3 & -2 & 3 \\ 2 & 1 & -1 \\ 4 & -3 & 2 \end{bmatrix}$ , then the matrix  $(\text{adj}A)A$  is equal to

1.  $17I$

2.  $-17I$

3.  $15I$

4.  $-15I$

Options 1. 1

2. 2

3. 3

4. 4

Q.84

For  $x > 1$ ,  $\int \frac{e^{7 \log x} - e^{5 \log x}}{e^{5 \log x} - e^{4 \log x}} dx$  equals.

1.  $\frac{x^3}{3} + \frac{x^2}{2} + C$  :  $C$  is a constant of integration

2.  $\frac{x^3}{3} - \frac{x^2}{2} + C$  :  $C$  is a constant of integration

3.  $\frac{x^3}{6} + \frac{x^2}{4} + C$  :  $C$  is a constant of integration

4.  $\frac{x^3}{6} - \frac{x^2}{4} + C$  :  $C$  is a constant of integration

Options 1. 1

2. 2

3. 3

4. 4

Q.85

The interval on which the function  $f(x) = x^4 - \frac{x^3}{3}$  is strictly decreasing, is:

1.  $(4, \infty)$

2.  $(\frac{1}{4}, \infty)$

3.  $(-\infty, \frac{1}{4})$

4.  $(0, \frac{1}{4})$

Options

1. 1

2. 2

3. 3

4. 4